

CO2-AT 1

Course Code: ITA0609

Course Title: Machine Learning

Student Name: S.Swagat

Reg No:192312007

1.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Given:

- **Objective Function:** $(f(x)=x^2)$
- **Initial Population:** {2, 4, 6, 8}

Step 1: Calculate Fitness Values

Fitness Function:

$$f(x)=x^2$$

Individual (x)	Fitness ($f(x)=x^2$)
2	4
4	16
6	36
8	64

Step 2: Select Best Candidates

Individuals with the **highest fitness values** are selected.

- **Selected Parents:** 8 and 6
- **Fitness Scores:** 64 and 36

Step 3: Binary Representation

Individual Binary (4-bit)

8 1000

6 0110

Step 4: Crossover (Exchange Least Significant Bit)

Parent 1: 1000

Parent 2: 0110

Exchange the **Least Significant Bit (LSB)**:

- **Offspring 1:** 1000 $\rightarrow$ 1000 = 8
- **Offspring 2:** 0110 $\rightarrow$ 0110 = 6

Since both LSBs are **0**, the offspring remain unchanged.

New Offspring: 8, 6

Step 5: Role of Mutation

Mutation randomly changes one or more bits in a chromosome with a small probability.

Example:

Offspring = **1000 (8)**

Mutate one bit:

1000 → 1001

New Individual = 9

Fitness:

$$f(9)=9^2=81$$

Since **81 > 64**, mutation has produced a **better solution**.

Final Result

- **Initial Population:** {2, 4, 6, 8}
- **Fitness Values:** {4, 16, 36, 64}
- **Selected Parents:** 8 and 6
- **Offspring after Crossover:** {8, 6}
- **Mutation Example:** 8 → 9
- **Improved Fitness after Mutation:** 81

Conclusion

- Fitness values are calculated using (**$f(x)=x^2$**).
- Individuals with the **highest fitness** are selected for reproduction.
- Crossover combines parent chromosomes to produce offspring.
- Mutation introduces random changes, increases genetic diversity, prevents premature convergence, and helps the Genetic Algorithm find better solutions.

2. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4 . For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

Given:

- **Weights:** $w_1 = 0.5$, $w_2 = 0.3$
- **Bias:** $b = -0.4$
- **Inputs:**
 - Advertisement Exposure (x_1) = 2
 - Income Level (x_2) = 1

Step 1: Calculate the Net Input

Formula:

$$z = (w_1 \times x_1) + (w_2 \times x_2) + b$$

Calculation:

$$z = (0.5 \times 2) + (0.3 \times 1) + (-0.4)$$

$$z = 1.0 + 0.3 - 0.4$$

$$z = 0.9$$

$$\text{Net Input} = 0.9$$

Step 2: Apply the Step Activation Function

Step Activation Rule:

- If $z \geq 0$, Output = 1
- If $z < 0$, Output = 0

Since $0.9 \geq 0$,

Output = 1

Final Decision

- **Net Input = 0.9**
- **Perceptron Output = 1**
- **Prediction: The customer will purchase the product.**

Conclusion

The perceptron computes a **net input of 0.9** using the given inputs, weights, and bias. Since the net input is **greater than or equal to zero**, the **step activation function produces an output of 1**. Therefore, the perceptron predicts that **the customer will purchase the product**. This demonstrates how a perceptron performs **binary classification** by making decisions based on weighted inputs.